



Recitation 8

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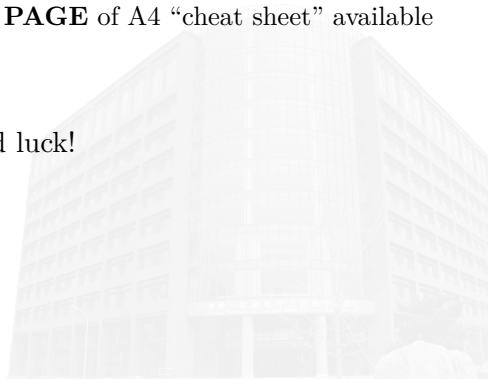
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Tsinghua University

April 14/15/16, 2021

Notice for Mid-term Exam

- Date: April 20, 2021 (Week 9)
- Time: 9:50-11:25 (GMT+08:00, Beijing time) (In-class)
- Location: **Lecture hall, 3-205, Rohm Building**
- Coverage: Lecture 1 to 7
- Additional information:
 - Closed-book exam, with one **PAGE** of A4 “cheat sheet” available
 - Calculator **NOT** necessary
- Account for 25% in final grade
- Please get well-prepared! Good luck!



1. Homework

2. Mid-term Review

3. Recitation



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1. Consider the following variant of Buffon's needle problem (Example 3.11), which was investigated by Laplace. A needle of length l is dropped on a plane surface that is partitioned in rectangles by horizontal lines that are a apart and vertical lines that are b apart. Suppose that the needle's length l satisfies $l < a$ and $l < b$. What is the expected number of rectangle sides crossed by the needle? What is the probability that the needle will cross at least one side of some rectangle?



Let A be the event that the needle will cross a horizontal line, and let B be the probability that it will cross a vertical line. The probability that at least one line will be crossed is

$$\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B)$$

We have to notice that here A and B are **not independent**, so $\mathbf{P}(A \cap B) \neq \mathbf{P}(A)\mathbf{P}(B)$. We have to calculate with joint PDF.

$$\begin{aligned}\mathbf{P}(A \cap B) &= \mathbf{P}(X \leq (l/2) \sin \Theta, Y \leq (l/2) \cos \Theta) \\ &= \iint_{x \leq (l/2) \sin \theta, y \leq (l/2) \cos \theta} f_{X,Y,\Theta}(x, y, \theta) dx dy d\theta \\ &= \frac{8}{\pi ab} \int_0^{\pi/2} \int_0^{(l/2) \cos \theta} \int_0^{(l/2) \sin \theta} dx dy d\theta \\ &= \frac{2l^2}{\pi ab} \int_0^{\pi/2} \cos \theta \sin \theta d\theta = \frac{l^2}{\pi ab}.\end{aligned}$$

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- Sample space
 - List of possible outcomes
 - Mutually exclusive & Collectively exhaustive
- Event
 - Subset of the sample space
 - A collection of possible outcomes
- Axioms of probability
 - Nonnegativity: $\mathbf{P}(A) \geq 0$
 - Normalization: $\mathbf{P}(\text{universe}) = 1$
 - Countable additivity:
 - If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$
 - If $A_1, A_2, \dots$ are disjoint events, then
$$\mathbf{P}(A_1 \cup A_2 \cup \dots) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \dots$$

- Conditional probability

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

- Multiplication rule

$$\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \cdot \mathbf{P}(B \mid A) \cdot \mathbf{P}(C \mid A \cap B)$$

- Total probability theorem

$$\mathbf{P}(B) = \sum_{i=1}^n \mathbf{P}(A_i \cap B) = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{P}(B \mid A_i)$$

- Bayes' rule

$$\mathbf{P}(A_i \mid B) = \frac{\mathbf{P}(A_i) \mathbf{P}(B \mid A_i)}{\mathbf{P}(B)} = \frac{\mathbf{P}(A_i) \mathbf{P}(B \mid A_i)}{\sum_{j=1}^n \mathbf{P}(A_j) \mathbf{P}(B \mid A_j)}$$

- Independence and conditional independence of two events

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$

$$\mathbf{P}(A \cap B \mid C) = \mathbf{P}(A \mid C) \cdot \mathbf{P}(B \mid C)$$

- Independence of a collection of events

$$\mathbf{P}(A_i \cap A_j \cap \cdots \cap A_q) = \mathbf{P}(A_i)\mathbf{P}(A_j) \cdots \mathbf{P}(A_q)$$

for **any** distinct indices $i, j, \dots, q$ chosen from $\{1, \dots, n\}$

- Counting results:
 - Permutations of n objects: $n!$
 - k -permutations of n objects: $n!/(n-k)!$
 - Combinations of k out of n objects:

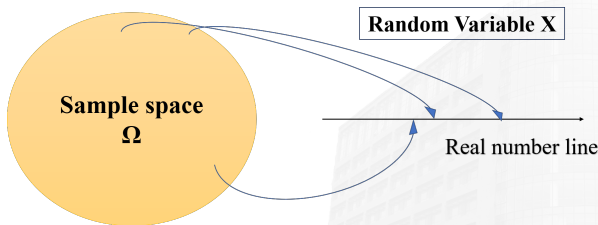
$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

- Partitions of n objects into r groups, with the i th group having n_i objects:

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \cdots n_r!}$$

- Random variables: A **function** from the sample space Ω to the real numbers
- PMF (for discrete random variables):

$$p_X(x) = \mathbf{P}(X = x) = \mathbf{P}\{\omega : X(\omega) = x\}$$



- Expectation

$$\mathbf{E}[X] = \sum_x x p_X(x)$$

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x)$$

$$\mathbf{E}[\alpha X + \beta] = \alpha \mathbf{E}[X] + \beta$$

- Variance

$$\begin{aligned}\text{var}(X) &= \mathbf{E}[(X - \mathbf{E}[X])^2] \\ &= \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2\end{aligned}$$

$$\text{var}(\alpha X + \beta) = \alpha^2 \text{var}(X)$$

- Conditional PMF

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A) = \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)}$$

- Conditional expectation

$$\mathbf{E}[X \mid A] = \sum_x x p_{X|A}(x)$$

$$\mathbf{E}[X \mid Y = y] = \sum_x x p_{X|Y}(x|y)$$

$$\mathbf{E}[g(X) \mid Y = y] = \sum_x g(x) p_{X|Y}(x|y)$$

- Total expectation theorem

$$\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X \mid A_1] + \cdots + \mathbf{P}(A_n)\mathbf{E}[X \mid A_n]$$

- Multiple discrete RVs

$$p_{X,Y}(x, y) = \mathbf{P}(X = x, Y = y) = \mathbf{P}(\{X(\omega) = x\} \cap \{Y(\omega) = y\})$$

- Expectation

$$\mathbf{E}[g(X, Y)] = \sum_x \sum_y g(x, y) p_{X,Y}(x, y)$$

- **Linearity property of expectation**

$$\mathbf{E}[X + Y + Z] = \mathbf{E}[X] + \mathbf{E}[Y] + \mathbf{E}[Z]$$

- **Holds for ANY RVs**
- Even the RVs are not independent!

- Independence and conditional independence

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \quad \forall x,y$$

$$p_{X,Y|A}(x,y) = p_{X|A}(x) \cdot p_{Y|A}(y) \quad \forall x,y$$

- If X and Y are independent:

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

- Continuous random variables

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx \quad \mathbf{P}(X \in A) = \int_A f_X(x) dx$$

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad \mathbf{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$\text{var}(X) = \mathbf{E}[(X - \mathbf{E}[X])^2] = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$$

- Cumulative distribution functions

$$F_X(x) = \mathbf{P}(X \leq x) = \begin{cases} \sum_{k \leq x} p_X(k), & \text{discrete} \\ \int_{-\infty}^x f_X(t) dt, & \text{continuous} \end{cases}$$

$$f_X(x) = \frac{dF_X}{dx}(x)$$

- Normal RV

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$$

- $\mathbf{E}[X] = \mu$, $\text{var}(X) = \sigma^2$
- $\frac{X - \mu}{\sigma} \sim N(0, 1)$
- LUT (Look Up Table) for CDF of standard normal

- Exponential RV

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- $\mathbf{E}[X] = \frac{1}{\lambda}$, $\text{var}(X) = \frac{1}{\lambda^2}$
- Memoryless property:

$$\mathbf{P}(X \geq c + x \mid X \geq c) = e^{-\lambda x} = \mathbf{P}(X \geq x)$$

- Joint PDF:

$$\mathbf{P}((X, Y) \in S) = \iint_S f_{X,Y}(x, y) dx dy$$

- Joint CDF:

$$F_{X,Y}(x, y) = \mathbf{P}(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(s, t) ds dt$$

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}}{\partial x \partial y}(x, y)$$

- Conditioning

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} \quad \text{if } f_Y(y) > 0$$

- The continuous Bayes' rule

$$f_{X|Y}(x | y) = \frac{f_X(x) f_{Y|X}(y | x)}{f_Y(y)} = \frac{f_X(x) f_{Y|X}(y | x)}{\int_{-\infty}^{\infty} f_X(t) f_{Y|X}(y | t) dt}$$

- Expectation:

$$\mathbf{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy$$

If g is linear, of the form $aX + bY + c$,

$$\mathbf{E}[aX + bY + c] = a\mathbf{E}[X] + b\mathbf{E}[Y] + c$$

- **Total expectation theorem:**

$$\mathbf{E}[X] = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{E}[X | A_i]$$

$$\mathbf{E}[X] = \mathbf{E}_Y [\mathbf{E}_{X|Y} [X | Y]] = \int_{-\infty}^{\infty} \mathbf{E}[X | Y = y] f_Y(y) dy$$

- Independence:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y), \quad \text{for all } x, y$$

$$f_{X|Y}(x | y) = f_X(x), \quad \text{for all } y \text{ with } f_Y(y) > 0 \text{ and all } x$$

- If X and Y are independent, then

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)] \quad \forall g, h$$

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

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1. In a row of $N + M$ seats N male students and M female students sit at random. Find the expected number of successive pairs of seats in which a male student and a female student sit.



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The probability p of a couple at the first seats is

$$\begin{aligned} p &= \frac{N}{N+M} \cdot \frac{M}{N+M-1} + \frac{M}{N+M} \cdot \frac{N}{N+M-1} \\ &= \frac{2NM}{(N+M)(N+M-1)} \end{aligned}$$

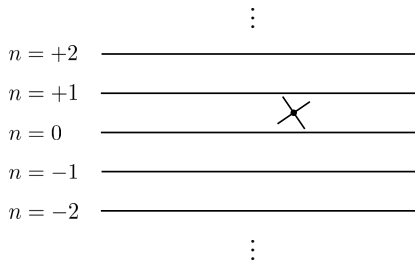
and the expected number of pairs at the first two seats is

$$p \cdot 1 + (1 - p) \cdot 0 = p.$$

The same is true for every pair of successive seats and there are $N + M - 1$ such pairs. The expected number of pairs is

$$p \cdot (N + M - 1) = \frac{2NM}{N + M}$$

2. The plane is ruled by the lines $y = n$, for $n = 0, \pm 1, \dots$, and on to this plane we drop a cross formed by welding together two unit needles perpendicularly at their midpoints. Let Z be the number of intersections of the cross with the grid of parallel lines. Find $\text{var}(Z)$.



We define the number of intersections of the first and the second needle as I and J , respectively. It can be first derived that $\mathbf{E}[I] = \mathbf{E}[J] = 2/\pi$ (from “Buffon’s needle”), followed by $\mathbf{E}[Z] = \mathbf{E}[I + J] = 4/\pi$.

Next we have

$$\begin{aligned}\text{var}(Z) &= \mathbf{E}[Z^2] - (\mathbf{E}[Z])^2 \\ &= \mathbf{E}[I^2 + J^2 + 2IJ] - (\mathbf{E}[Z])^2 \\ &= \frac{2}{\pi} + \frac{2}{\pi} + 2\mathbf{E}[IJ] - \frac{16}{\pi^2}.\end{aligned}$$

Since two intersections occurs when the distance l between the mid-point of the cross and its nearest line satisfies $l < \frac{1}{2} \min\{\sin \theta, \cos \theta\}$. Thus,

$$\begin{aligned}\mathbf{E}[IJ] &= \mathbf{P}(I = 1, J = 1) \\ &= \frac{4}{\pi} \int_0^{\frac{\pi}{2}} \int_0^{\frac{1}{2} \min\{\sin \theta, \cos \theta\}} dl d\theta \\ &= \frac{4}{\pi} \left(1 - \frac{\sqrt{2}}{2}\right)\end{aligned}$$

and hence

$$\begin{aligned}\text{var}(Z) &= \frac{4}{\pi} + 2 \cdot \frac{4}{\pi} \left(1 - \frac{\sqrt{2}}{2}\right) - \frac{16}{\pi^2} \\ &= \frac{12 - 4\sqrt{2}}{\pi} - \frac{16}{\pi^2}.\end{aligned}$$

3. The coronavirus, or COVID-19, has sickened more than a million people, forced millions to live under strict social distancing orders and paralyzed the global economy lately. Assume that you are visiting a city with 10^5 residents, among which any given individual may be infected by COVID-19 with probability 10^{-5} independently of all the others.

- (a) What is the probability that there is at least one other infected person given that the authority finds one?
- (b) What is the probability that there is at least one more if the authority finds two such people?
- (c) Based on (a) and (b), how many infected people need to be found before the authority can be reasonably confident that they have found them all, i.e., with probability r that there are no more infected people? Please give the implicit expression of the number of people.

(*Hint*: Consider Binomial and Poisson distributions)

- (a) The number X of such extraordinary individuals has the $\text{Bin}(10^5, 10^{-5})$ distribution. Hence $\mathbb{E}(X) = 1$ and

$$\begin{aligned}\mathbb{P}(X > 1 | X \geq 1) &= \frac{\mathbb{P}(X > 1)}{\mathbb{P}(X > 0)} = \frac{1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1)}{1 - \mathbb{P}(X = 0)} \\ &= \frac{1 - (1 - 10^{-5})^{10^5} - 10^5 \cdot 10^{-5} (1 - 10^{-5})^{10^5 - 1}}{1 - (1 - 10^{-5})^{10^5}} \\ &\simeq \frac{1 - 2e^{-1}}{1 - e^{-1}} \simeq 0.4\end{aligned}$$

- (b) Likewise

$$\mathbb{P}(X > 1 | X \geq 1) \simeq \frac{1 - 2e^{-1} - 0.5e^{-1}}{1 - 2e^{-1}} \simeq 0.3$$

(c) Provided $m \ll N = 10^5$

$$\mathbb{P}(X = m) = \frac{N!}{m!(N-m)!} \left(\frac{1}{N}\right)^m \left(1 - \frac{1}{N}\right)^{N-m} \simeq \frac{e^{-1}}{m!}$$

the Poisson distribution.

Assume that "reasonably confident that n is all" means that $\mathbb{P}(X > n | X \geq n) \leq r$ for some suitable small number r . Assuming the Poisson approximation, $\mathbb{P}(X > n) \leq r\mathbb{P}(X \geq n)$ iff.

$$e^{-1} \sum_{k=n+1}^{\infty} \frac{1}{k!} \leq r e^{-1} \sum_{k=n}^{\infty} \frac{1}{k!}$$

For any given r the smallest acceptable value of n may be determined numerically.

If r is small, then very roughly $n \simeq 1/r$ will do.

4. Suppose we choose three points independently at random on the circumference of a circle. Denote by $A(x)$ the probability that at least one of the angles of this triangle is larger than $x\pi$.

- (a) What is the value of $A(x)$ as $x \in [1/3, 1/2]$?
- (b) What is the value of $A(x)$ as $x \in [1/2, 1]$?
- (c) Calculate the density and expectation of the largest angle in the triangle.

(a&b) If O is the centre of the circle, take the radius OA as origin of coordinates. That is, $A = (1, 0)$, $B = (1, \theta)$, $C = (1, \phi)$, in polar coordinates, where we choose the labels in such a way that $0 \leq \theta \leq \phi$. $f(\theta, \phi) = (2\pi^2)^{-1}$ for $0 < \theta < \phi < 2\pi$.

The three angles of ABC are $\frac{1}{2}\theta$, $\frac{1}{2}(\phi - \theta)$, $\pi - \frac{1}{2}\phi$. You should plot in the θ/ϕ plane the set of pairs (θ, ϕ) such that $0 < \theta < \phi < 2\pi$ and such that at least one of the three angles exceeds $x\pi$. Then integrate f over this region to obtain the result. The shape of the region depends on whether or not $x < 1/2$.

$$A(x) = \begin{cases} 1 - (3x - 1)^2 & \text{if } \frac{1}{3} \leq x \leq \frac{1}{2} \\ 3(1 - x)^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

- (c) The density function g of the largest angle is given by differentiation:

$$g(x) = \begin{cases} 6(3x - 1) & \text{if } \frac{1}{3} \leq x \leq \frac{1}{2} \\ 6(1 - x) & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

The expectation is found to be $\frac{11}{18}\pi$.